

Please Download superstore data from Google and do the following questions.

Part 1: CASE Statement Questions

Write SQL queries to solve these:

1. Sales Performance Category

Create a column `SalesPerformance`:

- `'High'` if `Sales > 500`
- `'Medium'` if `Sales BETWEEN 200 AND 500`
- `'Low'` otherwise.

2. Profitability Flag

Add a column `ProfitFlag`:

- `'Profit'` if `Profit > 0`
- `'Loss'` if `Profit <= 0`.

3. Discount Label

Label orders based on `Discount`:

- `'No Discount'` if `Discount = 0`
- `'Low Discount'` if `Discount <= 0.2`
- `'High Discount'` otherwise.

4. Shipping Speed

Categorize `ShipMode`:

- `'Fast'` if `ShipMode IN ('Same Day', 'First Class')`
- `'Standard'` otherwise.

5. Segment Tag

Tag customer `Segment`:

- 'Corporate Client' for Segment='Corporate'
- 'Consumer Client' for Segment='Consumer'
- 'Home Office Client' otherwise.

6. Profit Range Buckets

Create a column ProfitRange:

- 'High Profit' if Profit > 100
- 'Medium Profit' if Profit BETWEEN 0 AND 100
- 'Loss' if Profit < 0.

7. Yearly Sales Tag

Create OrderYear from OrderDate and classify:

- 'Old Order' if year < 2017
- '2017 Order' if year = 2017
- 'Recent Order' otherwise.

8. Order Size

Classify based on Quantity:

- 'Bulk' if Quantity > 5
- 'Single Item' if Quantity = 1
- 'Small Order' otherwise.

9. Region Grouping

Group Region:

- 'East/West' if Region IN ('East','West')
- 'Central/South' otherwise.

10. Profit Margin Category

Assume $\text{ProfitMargin} = \text{Profit} / \text{Sales}$.

Categorize:

- 'Excellent' if margin > 0.3
- 'Good' if margin BETWEEN 0.1 AND 0.3
- 'Poor' otherwise.

11. Product Price Level

Bucket based on Sales per Quantity ($\text{Sales}/\text{Quantity}$):

- 'Premium' if > 200
- 'Midrange' if BETWEEN 50 AND 200
- 'Budget' otherwise.

12. High-Value Order Tag

Create a boolean column:

- 'Yes' if Sales > 1000
- 'No' otherwise.

13. Customer Loyalty Tier

Assign tiers based on total Sales per CustomerID:

- 'Platinum' if total sales > 5000
- 'Gold' if BETWEEN 3000 AND 5000
- 'Silver' otherwise.

14. State-Level Classification

Create a column StateType:

- 'CA State' if State = 'California'
- 'NY State' if State = 'New York'

- 'Other' otherwise.

15. Discount-Profit Interaction

Flag records:

- 'High Discount Loss' if `Discount > 0.3 AND Profit < 0`
- 'High Discount Profit' if `Discount > 0.3 AND Profit > 0`
- 'Regular' otherwise.

Part 2: Pivot-Style Questions with CASE Statements

Q1. Sales by Region as Columns

Create a pivot where each region becomes a column and show total sales

Q2. Quantity Sold by Segment

Pivot segments as columns and show total quantity.

Q3. Yearly Sales Pivot

Extract year from `OrderDate` and pivot:

Q4. ShipMode Count by Region

Count orders per `ShipMode` per region:

Q5. Profit by Category and Segment

Pivot Segment as columns, showing `Profit`: